Kristina P. Sinaga, Ph.D

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https://kpnaga08.github.io

Education

2016 – 2020 **Ph.D., Chung Yuan Christian University, Taiwan** in Applied Mathematics.
Thesis title: *Multi-View Fuzzy Clustering Algorithms for Multi-View Data*
Advisor: Prof. Miin-Shen Yang

2013 – 2015 **M.Sc., University of Sumatera Utara, Indonesia** in Mathematics.
Thesis title: *Model Optimasi Stokastik Penentuan Lokasi dan Jumlah Ambulan dengan Kolerasi (in Bahasa Indonesia)*
Advisor: Prof. Herman Mawengkang and Dr. Esther Nababan

2008 – 2013 **B.Sc., University of Sumatera Utara, Indonesia** in Mathematics.
Thesis title: *Analisis Pengaruh Produk Domestik Regional Bruto, Pendidikan dan Pengangguran terhadap Kemiskinan di Kabupaten/Kota Propinsi Sumatera Utara (in Bahasa Indonesia)*
Advisor: Prof. Tulus and Dr. Open Darnius

Areas of interest

My research interests include the computational intelligence, pattern recognition, feature learning, classification, and clustering. More broadly I am interested in mathematics behind Machine Learning and Deep Learning.

Experience

November 2020 – April 2022 **Lecturer** , Bina Nusantara University, Indonesia.

Septemer 2016 – August 2020 **PhD Research Student** , Chung Yuan Christian University, Taiwan.

Research Publications

Journal Articles

1. **Sinaga, K. P.**, Hussain, I. & Yang, M.-S. (2021). Entropy k-means clustering with feature reduction under unknown number of clusters. *IEEE Access*, *9*, 67736–67751.
2. Yang, M.-S. & **Sinaga, K. P.** (2021). Collaborative feature-weighted multi-view fuzzy c-means clustering. *Pattern Recognition*, 108064.
3. **Sinaga, K. P.** & Yang, M.-S. (2020). Unsupervised k-means clustering algorithm. *IEEE Access*.
4. Yang, M.-S. & **Sinaga, K. P.** (2019). A feature-reduction multi-view k-means clustering algorithm. *IEEE Access*, *7*, 114472–114486.
5. **Sinaga, K. P.**, Hutahaean, M. & Gea, P. (2016). Spatial variation in infant mortality with geographically weighted poisson regression (gwpr) approach. *vol*, *5*(3), 96–100.

6. **Sinaga, K. P.** (2015a). Model optimasi stokastik penentuan lokasi dan jumlah ambulan dengan kolerasi.
7. **Sinaga, K. P.** (2015b). Pemodelan data kemiskinan di provinsi sumatera utara dengan metode geographically weighted regression.
8. **Sinaga, K. P.** (2015c). Poverty data modeling in north sumatera province using geographically weighted regression (gwr) method. *International Journal of Science and Research (IJSR)*, *4*(2), 1738–1742.
9. **Sinaga, K. P.** & Tulus, O. D. (2013). Analisis pengaruh produk domestik regional bruto, pendidikan dan pengangguran terhadap kemiskinan di kab/kota propinsi sumatera utara tahun 2010–2011. *Saintia Matematika*, *1*(3), 261–271.

Conference Proceedings

1. Alexandra, J. & **Sinaga, K. P.** (2021). Machine learning approaches for marketing campaign in portuguese banks, In *2021 3rd international conference on cybernetics and intelligent system (icoris)*. IEEE.
2. Wibowo, H. & **Sinaga, K. P.** (2021). Telecommunication analytics based on customer segmentation using unsupervised algorithms, In *2021 3rd international conference on cybernetics and intelligent system (icoris)*. IEEE.

Books and Chapters

1. Yuniati, D. & **Sinaga, K. P.** (2021). *Analytics-based on classification and clustering methods for local community empowerment in indonesia*. In: Soft Computing in Data Science. Vol. 1489. CCIS. Springer, October 2021, pp. 133-145. https://doi.org/10.1007/978-981-16-7334-4_10
2. **Sinaga, K. P.**, Hsieh, J.-N., Benjamin, J. B. & Yang, M.-S. (2018). *Modified relational mountain clustering method*. In: Artificial Intelligence & Soft Computing. Vol. 10841. LNCS. Springer, May 2018, pp. 690-701. https://doi.org/10.1007/978-3-319-91253-0_64

Skills

Languages	Strong reading, writing and speaking competencies for English, Bahasa Indonesia.
Coding	MATLAB, R, LaTeX, Pyhton, MySQL, WEKA.
Misc.	Academic research, LaTeX typesetting.

Certifications and Licenses

February 2021	AWS-PDH IEEE Continuing Education
January 2021	UNSW-Learning to Teach Online

Workshop & Seminars

25 October 2021	**Presenter** , The 3rd International Conference on Cybernetics and Intelligent Systems (ICORIS), Virtually
17–18 March 2021	**Participant** , Southeast Asia Development Symposium 2021 (SEADS 2021), Online
4–18 November 2020	**Participant** , 2020 Mindanawan Math-Stat International Webcon, Via Zoom and Facebook Live

Workshop & Seminars (continued)

13 – 15 October 2019 — **Participant** , The 1 Asian International Conference in Science (AICS-2019), Chung Yuan Christian University, Taiwan.

3 – 7 June 2018 — **Poster Presenter** , June 2018 International Conference on Artificial Intelligence and Soft Computing (ICAISC 2018), Zakopane, Poland.

12 – 22 February 2017 — **Presenter** , Joint Seminar and Research Camp program on Mathmatical Science (JSRC2017), Niigata University, Japan.

28 – 29 November 2015 — **Presenter** , Seminar Nasional Matematika dan Terapan (SIMANTAP 2015), IAIN Zawiyah Cot Kala Langsa, Indonesia.

5 – 5 November 2015 — **Participant** , Seminar Nasional Analisis Matematika dan Terapannya, Departemen Matematika FMIPA Universitas Sumatera Utara, Medan, Indonesia.

29 – 29 August 2015 — **Presenter** , International Seminar on Operational Research (INTERIOR), Grand Aston, Medan, Indonesia.

28 – 29 November 2014 — **Presenter** , Seminar Nasional Matematika dan Terapan (SIMANTAP 2014), Universitas HKBP Nommensen, Pematang Siantar, Indonesia.

26 – 26 November 2014 — **Participant** , International Seminar of Operational Research and Its Role, Faculty of Mathematics and Natural Sciences, University of Sumatera Utara, Medan, Indonesia.

4 – 16 October 2014 — **Presenter** , ICMSA IMT GT 2014 , Terengganu, Malaysia.

4 March – 14 April 2011 — **Participant** , Pelatihan SPSS, Pusat Sistem Informasi Universitas Sumatera Utara, Medan, Indonesia.

27 – 27 November 2010 — **Participant** , Tantangan dan peluang Lulusan Perguruan Tinggi di Tengah Perkembangan Industri ICT Indonesia dan Dunia, Auditorium Universitas Sumatera Utara, Medan, Indonesia.

9 – 9 November 2010 — **Participant** , International Seminar on Mathematics and Its Application, University of Sumatera Utara, Medan, Indonesia.

28 – 28 November 2008 — **Participant** , Application of Statistics in Telecomunication, University of Sumatera Utara, Medan, Indonesia.

16 Juli – 28 September 2007 — **Participant** , Microsoft Office 2003 Proffesional, Pusat Pendidikan Komputer dan Internet WINET, Pematangsiantar, Indonesia.

Miscellaneous Experience

Awards and Achievements

2016 — **CYCU International Student Scholarship**, Chung Yuan Christian University, Taiwan.

2017 — **The recipient of the Japan Student Service Organization (JASSO)**, Joint Seminar and Research Camp program on Mathmatical Science (JSRC2017) between PKNU, MSU-IIT, NSU, KMUTT, NEU, CYCU, NSYSU, NCKU,NCUE, NTNU, NKNU, NTHU and Niigata University (NU) in Niigata University, Japan.

— **CYCU International Student Scholarship**, Chung Yuan Christian University, Taiwan.

2018 — **CYCU International Student Scholarship**, Chung Yuan Christian University, Taiwan.

Miscellaneous Experience (continued)

- **The Japan Science and Technology Agency (JST) recipient**, Activities of Sakura Science Exchange Program in Niigata University, Japan.

- **Support from Ministry of Science and Technology (MOST) Taiwan**, For International Conference on Artificial Intelligence and Soft Computing (ICAISC 2018), Zakopane, Poland, June 3-7, 2018.

2019
- **CYCU International Student Scholarship**, Chung Yuan Christian University, Taiwan.

- **Outstanding Doctoral Student Research Awards**, Chung Yuan Christian University, Taiwan.

2020
- **Honorary member of The Phi Tau Phi Scholastic Honor Society of The Republic of China.**

Professional Service

December 2020 – December 2021
- **Member** , IEEE

2020, 2021, 2022
- **Invited Reviewer** , IEEE Access

January 2022
- **Invited Reviewer** , IEEE Transactions on Knowledge and Data Engineering

- **Invited Reviewer** , Information Fusion, Elsevier

February 2022
- **Invited Reviewer** , Applied Soft Computing, Elsevier

March 2022
- **Invited Reviewer** , WCCI2022, IEEE WORLD CONGRESS ON COMPUTATIONAL INTELLIGENCE